

UCS310:Database Management Systems

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3	0	2	4.0

Course Objective:Emphasis is on the need of database systems. Main focus is on E-R diagrams, relational database, concepts of normalization and de-normalization and SQL commands.

Syllabus

Introduction: Data, data processing requirement, desirable characteristics of an ideal data processing system, traditional file-based system, its drawback, concept of data dependency, Definition of database, types of database, database management system, 3-schema architecture, database terminology, benefits of DBMS.

Relational Database: Relational data model: Introduction to relational database theory: definition of relation, keys, relational model integrity rules, introduction to Relational Algebra.

Database Analysis: Conceptual data modeling using E-R data model -entities, attributes, relationships, generalization, specialization, specifying constraints, Conversion of ER Models to Tables, Practical problems based on E-R data model.

Database Design: Functional Dependency, Canonical Covers, Candidate Key Identification, Normalization- 1NF, 2NF, 3NF, BCNF, 4NF and 5NF. Concept of De- normalization and practical problems based on these forms.

Transaction Management and Concurrency control: Concept of Transaction, States of Transaction and ACID properties, Need of Concurrency control, concept of Lock, Two phase locking protocol.

Recovery Management: Need of Recovery Management, Concept of Stable Storage, Log Based Recovery Mechanism, Checkpoint.

Database Implementation: Introduction to SQL, DDL aspect of SQL, DML aspect of SQL – update, insert, delete & various form of SELECT- simple, using special operators, aggregate functions, group by clause, sub query, joins, co-related sub query, union clause, View, exist operator. PL/SQL - cursor, stored function, stored procedure, triggers, error handling, and package.

Laboratory Work

Students will perform SQL commands to demonstrate the usage of DDL and DML, joining of tables, grouping of data and will implement PL/SQL constructs. They will also implement one project.

Project: It will contain database designing & implementation, should be given to group of 2-4 students. While doing projects emphasis should be more on back-end programming like use of SQL, concept of stored procedure, function, triggers, cursors, package etc. Project should have a continuous evaluation and should be spread over different components.

Healthcare data: including patient records, treatment statistics, and disease surveillance, can be managed in databases to improve healthcare services and outcomes.[3]

Construct a Databases that can manage educational resources, student performance data, and learning materials, helping ensure quality education is accessible and measurable.[4]

Data management systems can support urban planning, resource allocation, and sustainability efforts, helping to build smarter, more resilient cities.[11]

Course Learning Objectives (CLO)

The students will be able to:

1. Analyze the Information Systems as socio-technical systems, its need and advantages as compared to traditional file-based systems.
2. Analyze and design database using E-R data model by identifying entities, attributes and relationships.
3. Apply and create Relational Database Design process with Normalization and Denormalization of data.
4. Comprehend the concepts of transaction management, concurrence control and recovery management.
5. Demonstrate use of SQL and PL/SQL to implementation database applications.

Text Books

- . Database System Concepts, Silverschatz A., Korth F. H. and Sudarshan S., Tata McGraw Hill, 6th ed, 2010
- . Fundamentals of Database Systems, Elmasri R. and Navathe B. S., Pearson, 7th ed, 2016

Reference Books

1. SQL, PL/SQL the Programming Language of Oracle, Bayross I., BPB Publications, 4th ed, 2009
2. Modern Database Management, Hoffer J., Venkataraman, R. and Topi, H., Pearson, 12th ed 2016
3. Simplified Approach to DBMS, Parteek Bhatia and Gurvinder Singh,
4. Database management systems. Vol. 3. Raghu Ramakrishnan and Johannes Gehrke
5. FOR SQL/RA, New York: McGraw-Hill,